

Robust Online POMDP/MDP MCTS Planning with Anytime Deterministic Guarantees

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Abstract

Despite significant progress in robust planning, existing robust POMDP solvers remain offline, while online robust methods are restricted to fully observable MDPs and provide only probabilistic guarantees. We address this gap by introducing the first online anytime planner for robust POMDPs with deterministic guarantees. Our planner operates on a projected problem over sampled states and observations while deterministically bounding the probability mass of unsampled trajectories, yielding an anytime certificate on the approximation error. For any fixed policy, this certificate deterministically bounds its robust value under both rectangular and non-rectangular uncertainty sets. For rectangular uncertainty sets, applying the robust Bellman optimality operator further yields a deterministic upper bound on the optimal robust value. Together, these bounds guide online search and provide a certified stopping criterion for anytime planning, enabling early termination while guaranteeing optimality with respect to the original robust POMDP rather than its projected approximation. Building on these results, we present R-MCTS, the first online Monte Carlo tree search framework for robust POMDPs. We instantiate two variants: one using UCB exploration, which converges to the robust value of its induced tree policy, and another guided by our certified upper bound, which converges to the optimal robust value. Experimental results validate our theoretical guarantees, demonstrating that the certified intervals contain the true robust value and tighten as the sampled support grows.

Introduction

contributions:

1. A deterministic, anytime bound on the gap between the projected robust value of a fixed policy and its true robust value, $|V_0^{\pi,rob} - \hat{V}_0^{\pi,rob}| \leq \epsilon_0$, computable at no additional cost from the sampled tree; it holds at the root for any belief and, under exact belief updates, at every node.

2. Optimal-value upper bound. A deterministic robust upper bound on the optimal robust value, $V_{t,rect}^{*,rob}(b_t) \leq RUDB^{\pi}(b_t)$, obtained by applying the robust Bellman optimality operator to the certified upper bound.

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3. RUDB exploration. An exploration criterion that uses this upper bound to guide search, made valid at every node by decoupling belief update from robust backup — exact filtering with the nominal transition for validity, projected optimization for tractability — so the optimal-value certificate is available per-node without full-set optimization cost.

3. Both bounds are asymptotically exact: for any support-expanding online strategy $\hat{\mathcal{T}} \rightarrow \mathcal{T}$, the projected values converge to the true robust value $V^{\pi,rob}$ and, under the RUDB policy, to the optimal robust value $V_{rect}^{*,rob}$. The convergence rate is governed by the support-expansion rate, decoupling the guarantee from the specific solver.

4. R-MCTS framework and two solvers. A robust online planning framework that adds the above certificates to sampling-based search, instantiated as two solvers: (i) a POMCP-style planner with UCB exploration and robust projected backups, which inherits the fixed-policy certificate and converges to the robust value of the tree-policy; and (ii) an RUDB-exploration planner with exact (nominal) belief updates, which additionally attains the optimal-value guarantee and converges to the optimal robust value.

5. Certified early stopping and pruning. The two-sided bracket certifies action orderings during search: provably-suboptimal subtrees are pruned, and planning stops early once the optimal action is certified — rather than exhausting the sampling budget.

6. Post-hoc certification: the bound depends only on the evaluated policy and not on how it was produced, it certifies the robust value of any policy or planner, irrespective of how it was obtained.

8. A projection method that reduces a full RPOMDP to a tractable problem on the sampled support, under which robust backups optimize over projected sets rather than the full high-dimensional ones — and which provably preserves the worst-case value on that support.

Preliminaries

A robust POMDP accounts for model uncertainty by replacing the single transition and observation models with a set of admissible models. We consider a finite-horizon non-stationary RPOMDP $\mathcal{M}^{rob} =$

$(\mathcal{S}, \mathcal{A}, \mathcal{Z}, \mathcal{P}_{0:H-1}^T, \mathcal{P}_{1:H}^Z, b_0, r, H)$, where $\mathcal{S}$, $\mathcal{A}$, and $\mathcal{Z}$ are the finite state, action, and observation spaces. The sets $\mathcal{P}_{0:H-1}^T, \mathcal{P}_{1:H}^Z$ contain admissible non-stationary transition and observation model sequences $P_{0:H-1}^T, P_{1:H}^Z$, over the horizon H . The information available to the agent is summarized by the history of actions and observations up to time t which is denoted by $h_t \triangleq \{b_0, a_{0:t-1}, z_{1:t}\}$. This history induces a posterior distribution over the current hidden state, called the belief, defined by $b_t(s) \triangleq \Pr(s_t = s \mid h_t)$, b_0 being the initial prior belief. At each time step t after taking action a_t and observing z_{t+1} , the belief is updated with the Bayesian filter $b_{t+1}(s_{t+1} \mid h_t, a_t, z_{t+1}) = \eta_t \cdot P_Z(z_{t+1} \mid s_{t+1}) \sum_{s_t \in \mathcal{S}} P_T(s_{t+1} \mid s_t, a_t) b_t(s_t)$, where η_t is the normalizing constant. The reward function $r : \mathcal{B}(\mathcal{S}) \times \mathcal{A} \rightarrow [r_{\min}, r_{\max}]$ is assumed to be bounded and state-dependent, and is defined as $r(b, a) = \mathbb{E}_{s \sim b}[r(s, a)]$. A policy specifies how actions are selected based on the current belief. For a finite-horizon problem, a policy is a sequence $\pi = \{\pi_t\}_{t=0}^H$, where each π_t maps a belief b_t to a distribution over actions: $\pi_t(a_t \mid b_t) \triangleq \Pr(a_t = a \mid b_t)$.

Given a sequence of nominal models $\hat{P}_{0:H-1}^T, \hat{P}_{1:H}^Z$, such that the kernels at time step k are $\hat{P}_k^T : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$, $\hat{P}_k^Z : \mathcal{S} \rightarrow \Delta(\mathcal{Z})$, the non-rectangular uncertainty sets over the planning horizon H are defined as,

$$\mathcal{P}_{0:H-1}^T = \{P_{0:H-1}^T : D_T(P_{0:H-1}^T \parallel \hat{P}_{0:H-1}^T) \leq \rho^T\} \quad (1)$$

$$\mathcal{P}_{1:H}^Z = \{P_{1:H}^Z : D_Z(P_{1:H}^Z \parallel \hat{P}_{1:H}^Z) \leq \rho^Z\}, \quad (2)$$

where D_T, D_Z are some distance metrics, and ρ_T and ρ_Z are given uncertainty radiuses. The robust value function evaluates π under the worst-case admissible model, and is defined as:

$$V_t^{\pi, \text{rob}}(b_t) = \inf_{\substack{P_{t:H-1}^T \in \mathcal{P}_{t:H-1}^T \\ P_{t+1:H}^Z \in \mathcal{P}_{t+1:H}^Z}} V_t^\pi(b_t; P_{t:H-1}^T, P_{t+1:H}^Z), \quad (3)$$

where $V_t^\pi(b_t; P_{t:H-1}^T, P_{t+1:H}^Z)$ is the value function given a sequence of models. To reduce clutter, when clear from context, we omit the explicit dependence on $P_{t:H-1}^T$ and $P_{t+1:H}^Z$, and write $V_t^\pi(b_t; P_{t:H-1}^T, P_{t+1:H}^Z)$ as

$$V_t^\pi(b_t) = r(b_t, \pi_t) + \mathbb{E}_{z_{t+1:H}} \sum_{j=t+1}^H r(b_j, \pi_j). \quad (4)$$

The non-rectangular formulation in (1)-(3), forces the adversary to choose the model sequence jointly, so the admissibility of the model at one time may depend on choices at others. This coupling prevents a dynamic-programming decomposition of the Bellman backup, motivating the rectangular relaxation. The rectangular sets at time step t are defined as: [this paragraph can still be clarified and reduced]

$$\mathcal{P}_t^{T, \text{rect}} = \bigotimes_{\substack{s \in \mathcal{S} \\ a \in \mathcal{A}}} \mathcal{P}_t^T(s, a), \quad \mathcal{P}_{t+1}^{Z, \text{rect}} = \bigotimes_{s \in \mathcal{S}} \mathcal{P}_{t+1}^Z(s), \quad (5)$$

$$\mathcal{P}_t^T(s, a) = \{P_t^T(\cdot \mid s, a) : D(P_t^T(\cdot \mid s, a) \parallel \hat{P}_t^T(\cdot \mid s, a)) \leq \rho_{s,a}^T\},$$

$$\mathcal{P}_{t+1}^Z(s) = \{P_{t+1}^Z(\cdot \mid s) : D(P_{t+1}^Z(\cdot \mid s) \parallel \hat{P}_t^Z(\cdot \mid s)) \leq \rho_s^Z\},$$

Rectangularity is imposed at four levels. First, nature may choose a different worst-case model at each stage independently of its choices made at other stages. Second, within each stage, transition and observation uncertainty are independent. Third, row-rectangular, every transition/observation row representing the conditional distribution $P_t^T(\cdot \mid s_t, a_t) \in \Delta\mathcal{S}$, $P_{t+1}^Z(\cdot \mid s_{t+1}) \in \Delta\mathcal{Z}$, is chosen independently. Fourth, history-rectangular, different nodes of the same step in the tree are independent. This represents an adversary that may adapt its models choices locally rather than committing in advance to a fixed sequence of models.

Adapting (3) to the rectangular setting, we denote the robust rectangular value function, that admits robust DP:

$$V_{t, \text{rect}}^{\pi, \text{rob}}(b_t) = r(b_t, \pi_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^{T, \text{rect}} \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^{Z, \text{rect}}}} \mathbb{E}_{z_{t+1}}^{P_t^T, P_{t+1}^Z} [V_{t+1, \text{rect}}^{\pi, \text{rob}}(b_{t+1})]. \quad (6)$$

The rectangular relaxation comes at a cost of a more conservative robust value than the non-rectangular robust value.

Since $\mathcal{P}_{t:H} \subseteq \mathcal{P}_{t:H}^{\text{rect}}$, we get $V_{t, \text{rect}}^{\pi, \text{rob}}(b_t) \leq V_t^{\pi, \text{rob}}(b_t)$.

[might need to add a section in the Preliminaries on MCTS - let's decide later]

Certified Anytime Robust Online Solver

Computing the exact optimal robust value is typically infeasible in online POMDP planning, since the belief tree grows exponentially and each robust backup requires optimization over transition and observation uncertainty sets. We therefore plan over a projected belief tree containing only the sampled states and observations at each history/action node. The planner computes robust values on this projected problem, while the probability mass of trajectories leaving the sampled support is not ignored: it is bounded deterministically and used as an anytime certificate on the approximation error. We apply bellman optimality over the certified robust upper bounds which bounds the optimal rectangular value function. [not sure if we should also mention here that as sampled support increase the bounds converge to the optimal rect value]

For some time step t in the horizon, the state-support is the sampled set $\hat{\mathcal{S}}(h_t^-) \subseteq \mathcal{S}$, the observation support is $\hat{\mathcal{Z}}(h_t) \subseteq \mathcal{Z}$, and the action support is $\hat{\mathcal{A}}(h_t^-) \subseteq \mathcal{A}$. Let us define the set of tree-trajectories, consistent with the sampled states/observations/action support: [why do we not have also a subspace of the Action space?][why do we need $h_{t:H}$? can just use h_H , and then h_t will be a subsequence of it for any $t \in [1, H]$]

$$\begin{aligned} \hat{\tau}_{t:H}(h_H) = & \hat{\mathcal{S}}(h_t^-) \times \hat{\mathcal{A}}(h_t^-) \times \hat{\mathcal{S}}(h_{t+1}^-) \times \hat{\mathcal{Z}}(h_{t+1}) \times \dots \\ & \times \hat{\mathcal{A}}(h_H) \times \hat{\mathcal{S}}(h_H^-) \times \hat{\mathcal{Z}}(h_H), \end{aligned} \quad (7)$$

such that a single trajectory sample is denoted by $\tau_{t:H} \in \hat{\tau}_{t:H}(h_{t:H})$. Important nuance, if for any time step $k \in [t, H]$, $s_k \notin \hat{\mathcal{S}}(h_k^-)$ or $z_k \notin \hat{\mathcal{Z}}(h_k)$, then $\tau_{t:H} \notin \hat{\tau}_{t:H}$. [revisit paragraph] The sampled trajectory set identifies the branches represented in the planning tree, we now define the projected transition and observation models $\hat{P}_t^T, \hat{P}_{t+1}^Z$ to be sub-probabilities, defined only on their relative state/observation

support:

$$\hat{P}_t^T(\cdot|s_t, a_t; h_{t+1}^-) \triangleq P_t^T(\cdot|s_t, a_t), \forall s_{t+1} \in \hat{\mathcal{S}}(h_{t+1}^-), \quad (8)$$

$$\hat{P}_{t+1}^Z(\cdot|s_{t+1}; h_{t+1}) \triangleq P_{t+1}^Z(\cdot|s_{t+1}), \forall z_{t+1} \in \hat{\mathcal{Z}}(h_{t+1}), \quad (9)$$

where $\hat{P}_t^T \in \mathbb{R}^{|\hat{\mathcal{S}}(h_{t+1}^-)|}$, $\hat{P}_{t+1}^Z \in \mathbb{R}^{|\hat{\mathcal{Z}}(h_{t+1})|}$ [Alternatively: $|\hat{P}_t^T| = |\hat{\mathcal{S}}(h_{t+1}^-)|$, $|\hat{P}_{t+1}^Z| = |\hat{\mathcal{Z}}(h_{t+1})|$], meaning the projected models are over a much lower dimensionality, and P_t^T, P_{t+1}^Z belong to the full uncertainty sets.

NOT SURE I NEED THIS:

The projected prior belief $\hat{b}_0$, is the belief prior b_0 of the RPOMDP restricted to the state-support at the root node

$$\hat{b}_0(\cdot) \triangleq \begin{cases} b_0(s_0), & s_0 \in \hat{\mathcal{S}}(h_0^-), \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

The posterior belief $\hat{b}_t$ is updated using the projected models, $\hat{b}_t(s_t; \hat{\mathcal{S}}(h_t)) = \hat{P}_{t+1}^Z(z_t|s_t) \sum_{s_{t-1} \in \hat{\mathcal{S}}(h_t^-)} \hat{P}_t^T(s_t|s_{t-1}, a_{t-1}) \hat{b}(s_{t-1})$. Thus, the unnormalized projected models induce an unnormalized posterior belief.

Using these we define the projected value function:

$$\begin{aligned} \hat{V}_t^\pi(\hat{b}_t) &\triangleq r(\hat{b}_t, \pi_t) + \mathbb{E}_{z_{t+1} \sim \widehat{\text{Pr}}(\cdot|\hat{b}_t, \pi_t)} \left[\hat{V}_{t+1}^\pi(\hat{b}_{t+1}) \right], \\ &= \sum_{j=t}^{t+H} \left[\sum_{\tau_{t,j} \in \hat{\mathcal{T}}_{t,j}} \prod_{k=t}^{j-1} \hat{P}_k^T(s_{k+1}|\pi_k, s_k) \hat{P}_{k+1}^Z(z_{k+1}|s_{k+1}) \right. \\ &\quad \left. \times \hat{b}(s_t) r(s_j, a_j) \right]. \quad (11) \end{aligned}$$

Projected Uncertainty Set

[we'll need to adjust later the opening here] Our online planner is robust with respect to sampled subsets of the planning tree, which implies that the uncertainty around the projected models significantly differs from the uncertainty around the nominal models of the RPOMDP.

We introduce a new concept we call the *projected uncertainty set*. This set is defined as the set of **all** valid subprobability vectors such that **each** vector can be extended to a full probability distribution that lies within the original uncertainty set, namely:

$$\hat{\mathcal{P}}_{t:H-1}^T(h_{H-1}) = \left\{ \hat{P}_{t:H-1}^T : \exists P_{t:H-1}^{T, \text{out}} \text{ s.t. } \left(\hat{P}_{t:H-1}^T \cup P_{t:H-1}^{T, \text{out}} \right) \in \mathcal{P}_{t:H-1}^T \right\} \quad (12)$$

$$\hat{\mathcal{P}}_{t+1:H}^Z(h_H) = \left\{ \hat{P}_{t+1:H}^Z : \exists P_{t+1:H}^{Z, \text{out}} \text{ s.t. } \left(\hat{P}_{t+1:H}^Z \cup P_{t+1:H}^{Z, \text{out}} \right) \in \mathcal{P}_{t+1:H}^Z \right\} \quad (13)$$

where for each time step t in the horizon, $P_t^{T, \text{out}} \in \mathbb{R}^{|\mathcal{S}^{\text{out}}(h_{t+1}^-)|}$, $P_{t+1}^{Z, \text{out}} \in \mathbb{R}^{|\mathcal{Z}^{\text{out}}(h_{t+1})|}$ are the complementary models to their project counterparts, and their dimension is the size of the unsampled-support. These projected uncertainty sets catch the original uncertainty of the RPOMDP with respect to the sampled support.

[== stopped here ==]

Projected Robust POMDP

The projected RPOMDP, is defined as the tuple $\hat{\mathcal{M}}^{\text{rob}} = (\hat{\mathcal{S}}, \hat{\mathcal{A}}, \hat{\mathcal{Z}}, \hat{\mathcal{P}}_{t:H-1}^T, \hat{\mathcal{P}}_{t+1:H}^Z, b_0, r, H)$. Given some policy π , the **projected [estimated? need to decide]** robust value function is defined as the worst case models choices over the projected uncertainty set:

$$\hat{V}_t^{\pi, \text{rob}}(\hat{b}_t) = \inf_{\substack{\hat{P}_{t:H-1}^T \in \hat{\mathcal{P}}_{t:H-1}^T \\ \hat{P}_{t+1:H}^Z \in \hat{\mathcal{P}}_{t+1:H}^Z}} \hat{V}_t^\pi(\hat{b}_t; \hat{P}_{t:H-1}^T, \hat{P}_{t+1:H}^Z) \quad (14)$$

(14) is the same objective (3) the agent seeks to maximize, but evaluated only over valid trajectories represented in the sampled tree.

Anytime Deterministic Guarantees

This section analyzes the performance guarantees of our approach, relating the approximated robust value function (14) to the theoretical robust value (3):

[Thm1 is valid for any uncertainty set, right? Not necessarily even a ball uncertainty set (1). Maybe we should have a notation for a general non-rectangular uncertainty set]

Theorem 1. Let b_0 be the belief at time $t = 0$ and $V_0^{\pi, \text{rob}}(b_t)$ be the theoretical robust value function that is achieved by following the policy π . Let $\hat{b}_0$ be the projected prior (10), $\hat{V}_t^{\pi, \text{rob}}(\hat{b}_0)$ be the projected robust value function (14) following the same policy π , and $\mathcal{P}_{0:H-1}^T$ and $\mathcal{P}_{1:H}^Z$ [should be with hat?] the projected [+non-rectangular?] transition and observation uncertainty sets (12)-(13). For any policy π and horizon H , the difference between the theoretical and the projected robust value functions is bounded deterministically by:

$$\left| V_0^{\pi, \text{rob}}(b_0) - \hat{V}_0^{\pi, \text{rob}}(\hat{b}_0) \right| \leq \epsilon_0^\pi(\hat{b}_0) \triangleq \quad (15)$$

$$R_{\max} \sum_{j=0}^H \left(1 - \inf_{\substack{\hat{P}_{0:j-1}^T \in \hat{\mathcal{P}}_{0:j-1}^T \\ \hat{P}_{1:j}^Z \in \hat{\mathcal{P}}_{1:j}^Z}} \sum_{\tau_{0:j} \in \hat{\mathcal{T}}_{0:j}} \widehat{\text{Pr}}(\tau_{0:j}) \right), \quad (16)$$

where the trajectory likelihood $\widehat{\text{Pr}}(\tau_{0:j}) = \hat{b}_0(s_0) \prod_{k=0}^{j-1} \hat{P}_k^T(s_{k+1}|\pi_k, s_k) \hat{P}_{k+1}^Z(z_{k+1}|s_{k+1})$ is used to reduce clutter.

Corollary 1. If the **full state space is available [rephrase]**, and the belief states are updated exactly at every node h_t , then the bound of Theorem 1 holds at every node of the tree, namely for all t in the horizon H :

$$\left| V_t^{\pi, \text{rob}}(b_t) - \hat{V}_t^{\pi, \text{rob}}(\hat{b}_t) \right| \leq \epsilon_t^\pi(\hat{b}_t). \quad (17)$$

Corollary 2. For any policy π , as the sampled trajectories set increases $\hat{\mathcal{T}} \rightarrow \mathcal{T}$ the projected robust value function (14) converges to the theoretical robust value function (3):

$$\hat{V}_0^{\pi, \text{rob}}(\hat{b}_0) \xrightarrow{\hat{\mathcal{T}} \rightarrow \mathcal{T}} V_0^{\pi, \text{rob}}(b_0). \quad (18)$$

Crucially Theorem 1 makes no assumption on the structure of the uncertainty sets, and thus holds **verbatim** for rectangular uncertainty sets.

Instantiating it in the rectangular setting bounds the rectangular robust value at the root, $|V_{0, \text{rect}}^{\pi, \text{rob}}(b_0) - \widehat{V}_{0, \text{rect}}^{\pi, \text{rob}}(\widehat{b}_0)| \leq \epsilon_{0, \text{rect}}^{\pi}(\widehat{b}_0)$, which is the setting we adopt in the sequel: rectangularity admits a dynamic-programming decomposition of the robust Bellman backup, while the bound, computable without extra cost, provides an anytime deterministic certificate of policy quality.

By Corollary 2, this certificate is asymptotically exact: the interval collapses [\[converges\]](#) onto the true robust value as the sampled support grows $\widehat{\mathcal{T}} \rightarrow \mathcal{T}$. The rate of collapse [\[convergence rate\]](#) is governed by how quickly the algorithm expands its support, thus the guarantee tightens monotonically with planning effort, making it well suited to online planning. [\[you mention here: monotonicity, asymptotic convergence, and convergence rate - consider formulating explicitly as properties\]](#)

Two uses follow. [\[state each in a separate paragraph, can use bold text in the beginning\]](#) First, the bound yields an early-stopping criterion. It brackets each action value in the interval $[\widehat{Q}_{\text{rect}}^{\pi, \text{rob}} - \epsilon_{\text{rect}}^{\pi}(\widehat{b}, a), \widehat{Q}_{\text{rect}}^{\pi, \text{rob}} + \epsilon_{\text{rect}}^{\pi}(\widehat{b}, a)]$, where $\epsilon_{\text{rect}}^{\pi}(\widehat{b}, a)$ is the action-value error defined in (20). When two actions' intervals do not overlap, their ordering is certified: an action whose upper bound falls below another's lower bound is provably suboptimal, and its sub-tree can safely be pruned from further search, while an action whose lower bound exceeds every other upper bound is certified optimal. Our algorithm applies this test at the root, terminating early once the root action is certified optimal rather than exhausting the budget. Under exact belief updates (Corollary 1) the same test applies at every node, pruning certified-suboptimal sub-trees and eliminating redundant search. [\[the first one is related to the next section\]](#) Second, the bound depends on the evaluated policy alone, not on its origin; it therefore yields post-hoc guarantees on the robust value of any policy or planning algorithm, irrespective of how it was obtained.

Post-hoc guarantees cccc [\[todo\]](#)

Early stopping and suboptimal pruning cccc [\[todo\]](#)

Optimality Guarantees

Theorem 1 bounds the projected value of a fixed policy π from above and below. We now leverage this to bound the optimal robust value $V_{t, \text{rect}}^{*, \text{rob}}$ by applying the Bellman optimality operator to the robust upper deterministic bound (RUDB) we define as: [\[clarify this is in the setting of exact belief updates and a rectangular uncertainty set\]](#)

$$\text{RUDB}^{\pi}(b_t, a_t) \triangleq r(b_t, a_t) + \inf_{\substack{\widehat{P}_t^T \in \widehat{\mathcal{P}}_t^T \\ \widehat{P}_{t+1}^Z \in \widehat{\mathcal{P}}_{t+1}^Z}} \left\{ \mathbb{E}_{z_{t+1}|b_t, a_t}^{\widehat{P}_t^T, \widehat{P}_{t+1}^Z} \left[\widehat{V}_{t+1, \text{rect}}^{\pi, \text{RUDB}}(b_{t+1}) \right] + \epsilon_{t, \text{rect}}^{\pi, \widehat{P}_t^T, \widehat{P}_{t+1}^Z}(b_t, a_t) \right\}, \quad (19)$$

where the action-value error is defined by fixing action a_t

and following π in the time steps $j \in [t+1, H]$ that follow:

$$\epsilon_{t, \text{rect}}^{\pi, \widehat{P}_t^T, \widehat{P}_{t+1}^Z}(b_t, a_t) = R_{\max} \left(1 - \sum_{\substack{\tau_{t:t+1} \\ \in \widehat{\mathcal{T}}_{t:t+1}}} b_t(s_t) \widehat{P}_t^T(s_{t+1}|s_t, a_t) \times \widehat{P}_{t+1}^Z(z_{t+1}|s_{t+1}) \right) + \mathbb{E}_{z_{t+1}|b_t, a_t}^{\widehat{P}_t^T, \widehat{P}_{t+1}^Z} \left[\epsilon_{t+1, \text{rect}}^{\pi, \text{RUDB}}(b_{t+1}) \right]. \quad (20)$$

The RUDB superscript denotes the current-step model $\widehat{P}_t^{*T}, \widehat{P}_{t+1}^{*Z}$ that jointly minimize (19); the value term $\widehat{Q}_{t, \text{rect}}^{\pi, \text{RUDB}}$ and the error term $\epsilon_{t, \text{rect}}^{\pi, \text{RUDB}}$ are both evaluated at these shared minimizers, rather than optimized independently. This coupling is what allows the bound to be assembled from a single robust backup.

Theorem 2. Let b_t be the belief at time step t , $\dot{P}_t^T$ be the nominal transition model that is used to update the posterior belief $b_{t+1}(s_{t+1}|h_t^-, z_{t+1}; \dot{P}_t^T) = \Pr(s_{t+1}|h_t)$, called the nominal-belief at time step $t+1$. The optimal robust value function at time step t over rectangular uncertainty sets, is upper bounded by:

$$V_{t, \text{rect}}^{*, \text{rob}}(b_t) \leq \text{RUDB}^{\pi}(b_t) \triangleq \max_{a_t \in \mathcal{A}} \left\{ \text{RUDB}^{\pi}(b_t, a_t) \right\}, \quad (21)$$

such that the policy π comes from applying Bellman optimality operator over $\text{RUDB}^{\pi}(b_t, a_t)$:

$$\pi_t(b_t) = \arg \max_{a_t \in \mathcal{A}_t} \left\{ \widehat{Q}_{t, \text{rect}}^{\pi, \text{RUDB}}(b_t, a_t) + \epsilon_{t, \text{rect}}^{\pi, \text{RUDB}}(b_t, a_t) \right\}, \quad (22)$$

A subtlety underlies the use of Theorem 2 as an exploration criterion. For the RUDB to guide search, the error must be a valid bound at every node of the tree, not only the root—which requires the posterior belief at each node to be exact. We therefore update beliefs with the full nominal transition model $\dot{P}_t^T$ yielding an exact posterior at every history node. Crucially, this exactness is confined to the belief update: the robust backups still optimize over the projected uncertainty sets $\widehat{\mathcal{P}}_t^T, \widehat{\mathcal{P}}_{t+1}^Z$ rather than the full sets. We thus pay additional computation only in the filter, while the backups—which carry the robustness—remain cheap, as each infimum ranges over the sampled support alone. [\[where does the sampled state support come from?\]](#)

This yields two practical consequences. First, under exact belief updates, the two-sided bracket certifies action orderings at every node, not only the root: the pruning and early-stopping criteria of the previous section thus apply throughout the tree, eliminating redundant search.

Finally, as the sampled support expands, the guarantee becomes exact:

Corollary 3. Let b_t be the belief as time step t , and π^{RUDB} be the RUDB policy (22). As the sampled trajectories set increases $\widehat{\mathcal{T}} \rightarrow \mathcal{T}$, the projected robust value function [\[need to define it somewhere\]](#) converges to the optimal robust value function over rectangular uncertainty sets, namely:

$$\widehat{V}_{t, \text{rect}}^{\pi^{\text{RUDB}}, \text{rob}}(b_t) \xrightarrow{\widehat{\mathcal{T}} \rightarrow \mathcal{T}} V_{t, \text{rect}}^{*, \text{rob}}(b_t) \quad (23)$$

[\[need to add the proof in supplementary\]](#)

References

Supplementary

Value function derivation

In this section we explicitly derive the value function, that is used for the bounds, considering a planning horizon of H time steps, an initial belief $b_t = p(s_t | h_t)$, and a deterministic/stochastic policy $a_t \sim \pi_t(b_t)$. We restate the return, reward and value function for convenience:

$$G_t^\pi = \sum_{j=t}^H r(b_j, a_j), \quad (24)$$

$$r(b_t(h_t), a_t) = \mathbb{E}_{s_t \sim b_t} [r(s_t, a_t)] = \sum_{s_t} b_t(s_t) r(s_t, a_t), \quad (25)$$

$$V_t^\pi(b_t) = \mathbb{E}_{z_{t+1:j}}^\pi \left[\sum_{j=t}^H r(b_j, a_j) | b_t \right] = \sum_{j=t}^{t+H} \mathbb{E}_{z_{t+1:j}}^\pi [r(b_j, a_j) | b_t]. \quad (26)$$

b_j is the belief obtained after starting from b_t and observing the sequence $z_{t+1:j}$. Using (25) and (26):

$$V_t^\pi(b_t) = \sum_{j=t}^{t+H} \sum_{z_{t+1:j}} \sum_{s_j} P(z_{t+1:j} | b_t) b_j(s_j) r(s_j, a_j). \quad (27)$$

Applying chain rule we rewrite $P(z_{t+1:j} | b_t) b_j(s_j) = P(s_j, z_{t+1:j} | b_t)$ and marginalizing over the hidden states sequence:

$$P(s_j, z_{t+1:j} | b_t) = \sum_{s_{t:j-1}} P(s_{t:j}, z_{t+1:j} | b_t). \quad (28)$$

Assuming Markovian models, the joint probability of the state-observation trajectory can be written as:

$$P(s_{t:j}, z_{t+1:j} | b_t) = b_t(s_t) \prod_{k=t}^{j-1} P_k^T(s_{k+1} | s_k, a_k) P_{k+1}^Z(z_{k+1} | s_{k+1}). \quad (29)$$

We now plug the last 2 expressions into the value function and get the final expression for the value function:

$$V_t^\pi(b_t) = \sum_{j=t}^{t+H} \sum_{z_{t+1:j}} \sum_{s_{t:j}} \prod_{k=t}^{j-1} P_k^T(s_{k+1} | s_k, a_k) P_{k+1}^Z(z_{k+1} | s_{k+1}) b_t(s_t) r(s_j, a_j). \quad (30)$$

Similarly, using the projected models defined in (8), (9), and the restricted belief (10), the simplified value function is:

$$\hat{V}_t^\pi(\hat{b}_t) = \sum_{j=t}^{t+H} \sum_{z_{t+1:j} \in \hat{\mathcal{Z}}_{t+1:j}} \sum_{s_{t:j} \in \hat{\mathcal{S}}_{t:j}} \prod_{k=t}^{j-1} \hat{P}_k^T(s_{k+1} | \pi_k, s_k) \hat{P}_{k+1}^Z(z_{k+1} | s_{k+1}) \hat{b}(s_t) r(s_j, a_j). \quad (31)$$

Proof of Theorem 1

$$\mathcal{R}_{t:k-1}^T(P_{t:k-1}^T; h_{t:k-1}) = \begin{cases} P_{t:k-1}^T, & s_{t:k-1} \in \hat{\mathcal{S}}_{t:k-1}, \\ 0, & \text{otherwise.} \end{cases} \quad (32)$$

$$\mathcal{R}_{t+1:k}^Z(P_{t+1:k}^Z; h_{t+1:k}) = \begin{cases} P_{t+1:k}^Z, & s_{t+1} \in \hat{\mathcal{S}}_{t+1:k} \wedge z_{t+1} \in \hat{\mathcal{Z}}_{t+1:k}, \\ 0, & \text{otherwise.} \end{cases} \quad (33)$$

[move here all the Lemmas required for the proof, including the Value function representation - currently appears as "Derivation of the Value Function" (state as Lemma/Conjecture); can have the proofs of these lemmas later]

The following definitions are essential for the proof:

1. The set of all trajectories that is consistent with the theoretical planning tree:

$$\mathcal{T}_{t:H} = S_t \times A \times S_{t+1} \times Z_{t+1} \times \cdots \times A \times S_H \times Z_H \quad (34)$$

2. The sets of trajectories where at least 1 state/observation sample is not consistent with the sampled planning tree:

$$\mathcal{T}_{t:H}^{\text{out}} = \mathcal{T}_{t:H} / \hat{\mathcal{T}}_{t:H} \quad (35)$$

Theorem 1 (restated). Let b_0 be the belief at time $t = 0$ and $V_0^{\pi, \text{rob}}(b_t)$ be the theoretical robust value function that is achieved by following the policy π . Let $\hat{b}_0$ be the projected prior (10), $\hat{V}_t^{\pi, \text{rob}}(\hat{b}_0)$ be the projected robust value function (14) following the same policy π and $\mathcal{P}_{0:H-1}^T, \mathcal{P}_{1:H}^Z$ the projected transition and observation uncertainty sets (12), (13). For any policy π and horizon H , the difference between the theoretical and the projected robust value functions is bounded deterministically by:

$$\left| V_0^{\pi, \text{rob}}(b_0) - \hat{V}_0^{\pi, \text{rob}}(b_0) \right| \leq R_{\max} \sum_{j=0}^H \left(1 - \inf_{\substack{\hat{P}_{0:j-1}^T \in \hat{\mathcal{P}}_{0:j-1}^T \\ \hat{P}_{1:j}^Z \in \hat{\mathcal{P}}_{1:j}^Z}} \sum_{\tau_{0:j} \in \hat{\mathcal{T}}_{0:j}} \widehat{\Pr}(\tau_{0:j}) \right) = \epsilon_0^\pi(\hat{b}_0) \quad (36)$$

Proof. Using the above derivation (30), (11) for the theoretical/projected value function we denote the expected reward contribution of a single time step j of the horizon as $U_j^\pi(b_0)$, $\hat{U}_j^\pi(\hat{b}_0)$, such that:

$$U_j^\pi(b_0) \triangleq \mathbb{E}_{z_{1:j}} [r(b_0, a_0)] = \sum_{z_{1:j}} \sum_{s_{0:j}} \prod_{k=0}^{j-1} P_k^T(s_{k+1} | s_k, a_k) P_{k+1}^Z(z_{k+1} | s_{k+1}) b_0(s_0) r(s_j, a_j), \quad (37)$$

$$\hat{U}_j^\pi(\hat{b}_0) \triangleq \hat{\mathbb{E}}_{z_{1:j}} [r(\hat{b}_0, a_0)] = \sum_{z_{1:j} \in \hat{\mathcal{Z}}_{1:j}} \sum_{s_{0:j} \in \hat{\mathcal{S}}_{0:j}} \prod_{k=0}^{j-1} \hat{P}_k^T(s_{k+1} | \pi_k, s_k) \hat{P}_{k+1}^Z(z_{k+1} | s_{k+1}) \hat{b}_0(s_0) r(s_j, a_j). \quad (38)$$

We derive the bound for a single time step and extended later to full horizon. For a fixed reward index $j \in \{0, \dots, H\}$, we define the robust expected reward as:

$$U_j^{\pi, \text{rob}}(b_0) \triangleq \min_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} U_j^\pi(b_0; P_{0:j-1}^T, P_{1:j}^Z), \quad (39)$$

and the projected robust expected reward as:

$$\hat{U}_j^{\pi, \text{rob}}(\hat{b}_0) \triangleq \min_{\substack{\hat{P}_{0:j-1}^T \in \hat{\mathcal{P}}_{0:j-1}^T \\ \hat{P}_{1:j}^Z \in \hat{\mathcal{P}}_{1:j}^Z}} \hat{U}_j^\pi(\hat{b}_0; \hat{P}_{0:j-1}^T, \hat{P}_{1:j}^Z), \quad (40)$$

using Lemma 2 together with the restriction maps $\mathcal{R}_{0:j-1}^T$ and $\mathcal{R}_{1:j}^Z$ (32),(33) that project the full uncertainty set onto the projected uncertainty set, we get the following equality:

$$\hat{U}_j^{\pi, \text{rob}}(\hat{b}_0) = \min_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} \hat{U}_j^\pi(\hat{b}_0; \mathcal{R}_{0:j-1}^T(P_{0:j-1}^T), \mathcal{R}_{1:j}^Z(P_{1:j}^Z)). \quad (41)$$

The **robust difference of the theoretical with the projected [rephrase]**:

$$\begin{aligned} & \left| U_j^{\pi, \text{rob}}(b_0) - \hat{U}_j^{\pi, \text{rob}}(\hat{b}_0) \right| \\ &= \left| \min_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} U_j^\pi(b_0; P_{0:j-1}^T, P_{1:j}^Z) - \min_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} \hat{U}_j^\pi(\hat{b}_0; \mathcal{R}_{0:j-1}^T(P_{0:j-1}^T), \mathcal{R}_{1:j}^Z(P_{1:j}^Z)) \right| \end{aligned} \quad (42)$$

$$\leq \sup_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} \left| U_j^\pi(b_0; P_{0:j-1}^T, P_{1:j}^Z) - \hat{U}_j^\pi(\hat{b}_0; \mathcal{R}_{0:j-1}^T(P_{0:j-1}^T), \mathcal{R}_{1:j}^Z(P_{1:j}^Z)) \right| \quad (43)$$

For the last transition we used Lemma 1. To reduce clutter, we make use of the following shorthands: $\Pr(\tau_{0:j}) = b_0(s_0) \prod_{k=0}^{j-1} P_k^T(s_{k+1} | \pi_k, s_k) P_{k+1}^Z(z_{k+1} | s_{k+1})$, the sampled trajectory set $\hat{\mathcal{T}}_{0:j}$ defined in (7), its complementary set $\mathcal{T}_{0:j}^{\text{out}}$ defined in (35) and $\mathcal{R}(\Pr(\tau)) = \hat{b} \prod \mathcal{R}^T(P^T) \cdot \mathcal{R}^Z(P^Z)$:

$$\begin{aligned}
&= \sup_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} \left| \sum_{\tau_{0:j} \in \mathcal{T}_{0:j}} \Pr(\tau_{0:j}) r(s_j, a_j) - \sum_{\tau_{0:j} \in \mathcal{T}_{0:j}} \mathcal{R}_{0:j}(\widehat{\Pr}(\tau_{0:j})) r(s_j, a_j) \right| \\
&= \sup_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} \left| \sum_{\tau_{0:j} \in \widehat{\mathcal{T}}_{0:j}} \underbrace{\left(\Pr(\tau_{0:j}) - \mathcal{R}_{0:j}(\widehat{\Pr}(\tau_{0:j})) \right)}_{\equiv 0} r(s_j, a_j) + \sum_{\tau_{0:j}^{\text{out}} \in \mathcal{T}_{0:j}^{\text{out}}} \underbrace{\left(\Pr(\tau_{0:j}^{\text{out}}) - \mathcal{R}_{0:j}(\Pr(\tau_{0:j}^{\text{out}})) \right)}_{\equiv 0} r(s_j, a_j) \right|.
\end{aligned} \tag{44}$$

The underbraced terms in both summands equals zero by the definition of the Restricted mapping $\mathcal{R}$, therefore:

$$= \sup_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} \left| \sum_{\tau_{0:j}^{\text{out}} \in \mathcal{T}_{0:j}^{\text{out}}} \Pr(\tau_{0:j}^{\text{out}}) r(s_j, a_j) \right| \leq R_{max} \sup_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} \sum_{\tau_{0:j}^{\text{out}} \in \mathcal{T}_{0:j}^{\text{out}}} \Pr(\tau_{0:j}^{\text{out}}), \tag{45}$$

where we define $R_{max} = \max\{|r_{\min}|, |r_{\max}|\}$. Since $\Pr(\tau_{0:j})$ defines a full probability distribution over the full trajectory set $\mathcal{T}_{0:j}$, we have $\sum_{\tau_{0:j} \in \mathcal{T}_{0:j}} \Pr(\tau_{0:j}) \equiv 1$. Therefore:

$$\sum_{\tau_{0:j}^{\text{out}} \in \mathcal{T}_{0:j}^{\text{out}}} \Pr(\tau_{0:j}^{\text{out}}) = \Pr(\mathcal{T}_{0:j}^{\text{out}}) = 1 - \widehat{\Pr}(\widehat{\mathcal{T}}_{0:j}) = 1 - \mathcal{R}_{0:j}(\widehat{\Pr}(\widehat{\mathcal{T}}_{0:j})), \tag{46}$$

and (45) equals:

$$R_{max} \sup_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} \left(1 - \mathcal{R}_{0:j}(\widehat{\Pr}(\widehat{\mathcal{T}}_{0:j})) \right) = R_{max} \left(1 - \inf_{\substack{P_{0:j-1}^T \in \mathcal{P}_{0:j-1}^T \\ P_{1:j}^Z \in \mathcal{P}_{1:j}^Z}} \mathcal{R}_{0:j}(\widehat{\Pr}(\widehat{\mathcal{T}}_{0:j})) \right), \tag{47}$$

making use again with Lemma 2:

$$= R_{max} \left(1 - \inf_{\substack{\widehat{P}_{0:j-1}^T \in \widehat{\mathcal{P}}_{0:j-1}^T \\ \widehat{P}_{1:j}^Z \in \widehat{\mathcal{P}}_{1:j}^Z}} \sum_{z_{1:j} \in \widehat{\mathcal{Z}}_{1:j}} \sum_{s_{0:j} \in \widehat{\mathcal{S}}_{0:j}} \widehat{b}_0(s_0) \prod_{k=0}^{j-1} \widehat{P}_k^T(s_{k+1} | \pi_k, s_k) \widehat{P}_{k+1}^Z(z_{k+1} | s_{k+1}) \right). \tag{48}$$

Extending to full horizon we get the final bound:

$$\left| V_0^{\pi, \text{rob}}(b_0) - \widehat{V}_0^{\pi, \text{rob}}(\widehat{b}_0) \right| = \left| \sum_{j=t}^H \left(U_j^{\pi, \text{rob}}(b_0) - \widehat{U}_j^{\pi, \text{rob}}(\widehat{b}_0) \right) \right| \leq \sum_{j=t}^H \left| U_j^{\pi, \text{rob}}(b_0) - \widehat{U}_j^{\pi, \text{rob}}(\widehat{b}_0) \right| \tag{49}$$

$$\leq R_{max} \sum_{j=0}^H \left(1 - \inf_{\substack{\widehat{P}_{0:j-1}^T \in \widehat{\mathcal{P}}_{0:j-1}^T \\ \widehat{P}_{1:j}^Z \in \widehat{\mathcal{P}}_{1:j}^Z}} \sum_{z_{1:j} \in \widehat{\mathcal{Z}}_{1:j}} \sum_{s_{0:j} \in \widehat{\mathcal{S}}_{0:j}} \widehat{b}_0(s_0) \prod_{k=0}^{j-1} \widehat{P}_k^T(s_{k+1} | \pi_k, s_k) \widehat{P}_{k+1}^Z(z_{k+1} | s_{k+1}) \right) \tag{50}$$

$$\triangleq \epsilon_0^{\pi}(\widehat{b}_0) \tag{51}$$

□

Lemma 1 Proof

Lemma 1. for any 2 function $f(\theta), \widehat{f}(\theta)$, the following holds:

$$\left| \min_{\theta} f(\theta) - \min_{\theta} \widehat{f}(\theta) \right| \leq \max_{\theta} |f(\theta) - \widehat{f}(\theta)| \tag{52}$$

Proof. Let

$$\varepsilon = \sup_{\theta \in \Theta} |f(\theta) - g(\theta)|$$

Then for every θ ,

$$f(\theta) \leq g(\theta) + \varepsilon$$

Taking minima gives

$$\min_{\theta} f(\theta) \leq \min_{\theta} g(\theta) + \varepsilon$$

Similarly, since

$$g(\theta) \leq f(\theta) + \varepsilon,$$

we get

$$\min_{\theta} g(\theta) \leq \min_{\theta} f(\theta) + \varepsilon$$

Therefore,

$$\left| \min_{\theta} f(\theta) - \min_{\theta} g(\theta) \right| \leq \varepsilon = \sup_{\theta \in \Theta} |f(\theta) - g(\theta)|$$

□

Lemma 2 Proof

Lemma 2. Let $\mathcal{P}$ denote some uncertainty set, and $\widehat{\mathcal{P}}$ denote its projected version. Let $\mathcal{R}$ be the restriction map that takes a full probability distribution $P \in \mathcal{P}$ and retains only the entries associated with the sampled support:

$$\mathcal{R}(P; \widehat{\mathcal{S}}) = \begin{cases} P(s), & s \in \widehat{\mathcal{S}}, \\ 0, & \text{otherwise.} \end{cases} \quad (53)$$

Then minimizing some function $f : \widehat{\mathcal{P}} \rightarrow \mathbb{R}$ over $\widehat{\mathcal{P}}$ is the same as minimizing $f(\mathcal{R}(P))$ over the full set $\mathcal{P}$, namely:

$$\inf_{P \in \mathcal{P}} f(\mathcal{R}(P)) = \inf_{\widehat{P} \in \widehat{\mathcal{P}}} f(\widehat{P}), \quad (54)$$

Proof. By the definition of the projected uncertainty sets as the images of the full uncertainty sets under the restriction maps, we have

$$\widehat{\mathcal{P}} = \mathcal{R}(\mathcal{P}), \quad (55)$$

We prove the desired equality by showing both inequalities. First, let

$$P \in \mathcal{P}, \quad (56)$$

be arbitrary full distribution. By definition:

$$\mathcal{R}(P) \in \widehat{\mathcal{P}}, \quad (57)$$

Therefore,

$$\inf_{\widehat{P} \in \widehat{\mathcal{P}}} f(\widehat{P}) \leq f(\mathcal{R}(P)). \quad (58)$$

Since this holds for every

$$P \in \mathcal{P}, \quad (59)$$

we can take the infimum over the full uncertainty set and obtain

$$\inf_{\widehat{P} \in \widehat{\mathcal{P}}} f(\widehat{P}) \leq \inf_{P \in \mathcal{P}} f(\mathcal{R}(P)). \quad (60)$$

For the opposite direction, let

$$\widehat{P} \in \widehat{\mathcal{P}}, \quad (61)$$

be arbitrary projected distribution. Since the projected uncertainty set is defined as the image of the full uncertainty set, there exist a full model:

$$P \in \mathcal{P}, \quad (62)$$

such that

$$\hat{P} = \mathcal{R}(P), \quad (63)$$

Therefore,

$$f(\hat{P}) = f(\mathcal{R}(P)) \quad (64)$$

Hence,

$$\inf_{P \in \mathcal{P}} f(\mathcal{R}(P)) \leq f(\hat{P}). \quad (65)$$

Since this holds for every

$$\hat{P} \in \hat{\mathcal{P}}, \quad (66)$$

we can take the infimum over the projected uncertainty set and obtain

$$\inf_{P \in \mathcal{P}} f(\mathcal{R}(P)) \leq \inf_{\hat{P} \in \hat{\mathcal{P}}} f(\hat{P}). \quad (67)$$

Combining the two inequalities gives equality:

$$\inf_{P \in \mathcal{P}} f(\mathcal{R}(P)) = \inf_{\hat{P} \in \hat{\mathcal{P}}} f(\hat{P}). \quad (68)$$

Which proves the Lemma. $\square$

Corollary 1 Proof

Corollary 1 (restated). *If the full state space is available, and the belief states are updated exactly at every node h_t of the tree then the bound of Theorem 1 holds at every node of the tree, namely for all t in the horizon H :*

$$\left| V_t^{\pi, \text{rob}}(b_t) - \hat{V}_t^{\pi, \text{rob}}(b_t) \right| \leq \epsilon_t^\pi(b_t) \quad (69)$$

Proof. The proof of theorem 1 relies on normalized belief in (46), namely:

$$\sum_{\tau_{t:j} \in \mathcal{T}_{t:j}} \Pr(\tau_{t:j}) = \sum_{z_{t+1:j}} \sum_{s_{t:j}} b_t(s_t) \prod_{k=t}^{j-1} P_k^T(s_{k+1} | \pi_k, s_k) P_{k+1}^Z(z_{k+1} | s_{k+1}) \equiv 1, \quad (70)$$

which requires only that the belief at the node be a true normalized distribution. Under exact belief updates this holds at every history node h_t and the theorem applies verbatim with $b_0, t=0$ replaced by b_t, t . $\square$

Theorem 2 Proof:

We restate for convenience the definitions of $RUDB^\pi(b_t, a_t)$ (19), and its derivative terms:

$$RUDB^\pi(b_t, a_t) \triangleq r(b_t, a_t) + \inf_{\substack{\hat{P}_t^T \in \hat{\mathcal{P}}_t^T \\ \hat{P}_{t+1}^Z \in \hat{\mathcal{P}}_{t+1}^Z}} \left\{ \mathbb{E}_{z_{t+1}|b_t, a_t}^{\hat{P}_t^T, \hat{P}_{t+1}^Z} \left[\hat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1}) \right] + \epsilon_{t, rect}^{\pi, \hat{P}_t^T, \hat{P}_{t+1}^Z}(b_t, a_t) \right\}, \quad (71)$$

$$= \hat{Q}_{t, rect}^{\pi, RUDB}(b_t, a_t) + \epsilon_{t, rect}^{\pi, RUDB}(b_t, a_t). \quad (72)$$

IMPORTANT: the RUDB superscript denotes the **shared** optimized models $\hat{P}_t^{*T}, \hat{P}_{t+1}^{*Z}$ computed by the infimum, meaning:

$$\hat{P}_t^{*T}, \hat{P}_{t+1}^{*Z} = \arg \inf_{\substack{\hat{P}_t^T \in \hat{\mathcal{P}}_t^T \\ \hat{P}_{t+1}^Z \in \hat{\mathcal{P}}_{t+1}^Z}} \left\{ \mathbb{E}_{z_{t+1}|b_t, a_t}^{\hat{P}_t^T, \hat{P}_{t+1}^Z} \left[\hat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1}) \right] + \epsilon_{t, rect}^{\pi, \hat{P}_t^T, \hat{P}_{t+1}^Z}(b_t, a_t) \right\}, \quad (73)$$

$$\hat{Q}_{t, rect}^{\pi, RUDB}(b_t, a_t) \triangleq r(b_t, a_t) + \mathbb{E}_{z_{t+1}|b_t, a_t}^{\hat{P}_t^{*T}, \hat{P}_{t+1}^{*Z}} \left[\hat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1}) \right], \quad (74)$$

$$\epsilon_{t, rect}^{\pi, RUDB}(b_t, a_t) = R_{max} \left(1 - \sum_{\tau_{t:t+1} \in \hat{\mathcal{T}}_{t:t+1}} b_t(s_t) \hat{P}_t^{*T}(s_{t+1}|s_t, a_t) \hat{P}_{t+1}^{*Z}(z_{t+1}|s_{t+1}) \right) + \mathbb{E}_{z_{t+1}|b_t, a_t}^{\hat{P}_t^{*T}, \hat{P}_{t+1}^{*Z}} \left[\epsilon_{t+1, rect}^{\pi, RUDB}(b_{t+1}) \right]. \quad (75)$$

Theorem 2 (restated). Let b_t be the belief at time step t , $\hat{P}_t^T$ be the nominal transition model that is used to update the posterior belief $b_{t+1}(s_{t+1}|h_t^-, z_{t+1}; \hat{P}_t^T) = \Pr(s_{t+1}|h_t)$, called the nominal-belief at time step $t + 1$. The optimal robust value function at time step t over rectangular uncertainty sets, is upper bounded by:

$$V_{t,rect}^{*,rob}(b_t) \leq RUDB^\pi(b_t) \triangleq \max_{a_t \in \mathcal{A}} \{RUDB^\pi(b_t, a_t)\}, \quad (76)$$

such that the policy π comes from applying bellman optimality operator over $RUDB^\pi(b_t, a_t)$:

$$\pi_t(b_t) = \arg \max_{a_t \in \mathcal{A}_t} \left\{ \hat{Q}_{t,rect}^{\pi, RUDB}(b_t, a_t) + \epsilon_{t,rect}^{\pi, RUDB}(b_t, a_t) \right\}, \quad (77)$$

Proof. We prove by induction, that by applying the bellman optimality operator on upper bounds to the robust value function, we get an upper bound on the **optimal** value function.

Base case ($t = H$): We set the terminal action-value to be the immediate expected reward:

$$Q_{H,rect}^{*,rob}(b_H, a_H) = \hat{Q}_{H,rect}^{\pi,rob}(b_H, a_H) = r(b_H, a_H) = \sum_{s_H \in \mathcal{S}} b_H(s_H) r(s_H, a_H). \quad (78)$$

At the terminal time step H of the POMDP, there is no future transition/observation term, therefore $\epsilon_{H,rect}^\pi(b_H, a_H) = 0$. Therefore the terminal $RUDB^\pi(b_H, a_H)$ on the action-value is:

$$RUDB^\pi(b_H, a_H) = r(b_H, a_H) \quad (79)$$

and the base-case for the induction is the best immediate expected reward:

$$V_{H,rect}^{*,rob}(b_H) \leq RUDB^\pi(b_H) \triangleq \max_{a_H \in \mathcal{A}_H} RUDB^\pi(b_H, a_H). \quad (80)$$

Induction Hypothesis: Given time step $t + 1$, assume that for any future sampled-observation $z_{t+1} \in \hat{\mathcal{Z}}_{t+1}(h_{t+1})$ and any nominal-belief $b_{t+1}(s_{t+1}|h_t^-, z_{t+1}; \hat{P}_t^T)$:

$$V_{t+1,rect}^{*,rob}(b_{t+1}) \leq RUDB^\pi(b_{t+1}). \quad (81)$$

This means that at the next depth of the tree $RUDB^\pi(b_{t+1})$ is already a valid upper bound on the optimal rectangular robust value.

Induction Step: We will now show that the hypothesis holds for time step t , specifically:

$$Q_{t,rect}^{*,rob}(b_t, a_t) \leq RUDB^\pi(b_t, a_t). \quad (82)$$

Writing the above LHS recursively:

$$Q_{t,rect}^{*,rob}(b_t, a_t) = r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} \left[V_{t+1,rect}^{*,rob}(b_{t+1}) \right], \quad (83)$$

by the induction hypothesis for any nominal-belief b_{t+1} at time step $t + 1$:

$$\leq r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} [RUDB^\pi(b_{t+1})],$$

following Lemma 3:

$$\leq \hat{Q}_{t,rect}^{\pi, RUDB}(b_t, a_t) + \epsilon_{t,rect}^{\pi, RUDB}(b_t, a_t) \triangleq RUDB^\pi(b_t, a_t), \quad (84)$$

since:

$$V_{t,rect}^{*,rob}(b_t) = \max_{a_t \in \mathcal{A}_t} Q_{t,rect}^{*,rob}(b_t, a_t) \leq \max_{a_t \in \mathcal{A}_t} RUDB^\pi(b_t, a_t) \triangleq RUDB^\pi(b_t), \quad (85)$$

we proved the induction step, and the theorem. $\square$

Lemma 3 Proof

Lemma 3. Let b_t be the nominal-belief at time step t , and b_{t+1} be the nominal-belief at the following time step $t + 1$. For any action $a_t \in \mathcal{A}_t$, and future action a_{t+1} selected by the RUDB-greedy policy $\pi_{t+1}(b_{t+1}) = \arg \max_{a_{t+1} \in \mathcal{A}_{t+1}} \{RUDB^\pi(b_{t+1}, a_{t+1})\}$, the following inequality holds:

$$r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} [RUDB^\pi(b_{t+1})] \leq RUDB^\pi(b_t, a_t).$$

Proof. By the definition of the $RUDB^\pi(b_{t+1}, a_{t+1})$ (19), at time step $t + 1$:

$$\begin{aligned} & r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} [RUDB^\pi(b_{t+1})] \\ &= r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} \left[\max_{a_{t+1} \in \mathcal{A}_t} \left\{ \hat{Q}_{t+1, rect}^{\pi, RUDB}(b_{t+1}, a_{t+1}) + \epsilon_{t+1, rect}^{\pi, RUDB}(b_{t+1}, a_{t+1}) \right\} \right]. \end{aligned} \quad (86)$$

The action a_{t+1} is selected after observing z_{t+1} , meaning after reaching the next history $h_{t+1} = \{h_t, a_t, z_{t+1}\}$. Therefore, for each posterior nominal-belief $b_{t+1}|z_{t+1}$, there is a corresponding action that maximizes (86):

$$a_{t+1}^{RUDB} = \pi_{t+1}(b_{t+1}|z_{t+1}) = \arg \max_{a_{t+1} \in \mathcal{A}_t} \{ \hat{Q}_{t+1, rect}^{\pi, RUDB}(b_{t+1}, a_{t+1}) + \epsilon_{t+1, rect}^{\pi, RUDB}(b_{t+1}, a_{t+1}) \} \quad (87)$$

[suggest to replace below a_{t+1}^{RUDB} with $\pi_{t+1}(b_{t+1})$; consider switching to $\pi_{t+1}^{RUDB*}(b_{t+1})$ (if so, in all places)]

$$= r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} \left[\hat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1}) + \epsilon_{t+1, rect}^{\pi, RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) \right], \quad (88)$$

[should i mention the switch: $\hat{Q}_{t+1, rect}^{\pi, RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) \rightarrow \hat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1})$???]

by the linearity of the expectation operator $\mathbb{E}[\cdot]$:

$$= r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \left\{ \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} \left[\hat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1}) \right] + \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} \left[\epsilon_{t+1, rect}^{\pi, RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) \right] \right\}. \quad (89)$$

We first focus on the **left** expectation from (89), and show that $\mathbb{E} \left[\hat{V}_{t+1, rect}^{\pi, RUDB} \right] = \hat{\mathbb{E}} \left[\hat{V}_{t+1, rect}^{\pi, RUDB} \right]$. The projected value function as derived in (11) is:

$$\hat{V}_t^\pi(b_t) = \sum_{j=t}^H \sum_{z_{t+1:j} \in \hat{\mathcal{Z}}_{t+1:j}} \sum_{s_{t:j} \in \hat{\mathcal{S}}_{t:j}} \prod_{k=t}^{j-1} \hat{P}_k^T(s_{k+1}|\pi_k, s_k) \hat{P}_{k+1}^Z(z_{k+1}|s_{k+1}) b(s_t) r(s_j, a_j) \quad (90)$$

As defined in (8), (9), the projected models $\hat{P}_t^T, \hat{P}_{t+1}^Z$ assign zero probability to any trajectory leaving the sampled support $\tau_{t:j} \notin \hat{\mathcal{T}}_{t:j}$, namely when at least 1 state/observation sample is outside the sampled support, which induces the projected value to be zero on such trajectories:

$$\hat{V}_t^\pi(b_t | \tau_{t:j} \notin \hat{\mathcal{T}}_{t:j}) \equiv 0, \quad \forall \tau_{t:j} \notin \hat{\mathcal{T}}_{t:j}, \quad \forall j \in [t, H] \quad (91)$$

writing the expectation $\mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} \left[\hat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1}) \right]$ explicitly:

$$\begin{aligned} & \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} \left[\hat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1}) \right] = \sum_{z_{t+1} \in \mathcal{Z}_{t+1}} \Pr(z_{t+1}|b_t, a_t) \hat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1}) \\ &= \sum_{\tau_{t:t+1} \in \hat{\mathcal{T}}_{t:t+1}} b_t(s_t) P_t^T(s_{t+1} | s_t, a_t) P_{t+1}^Z(z_{t+1} | s_{t+1}) \hat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1}), \end{aligned} \quad (92)$$

we split the sums to in/out of sampled-support, where $\tau_{t:t+1} = \{s_t, s_{t+1}, z_{t+1}\}$:

$$\begin{aligned}
&= \sum_{\tau_{t:t+1} \in \widehat{\mathcal{T}}_{t:t+1}} b_t(s_t) P_t^T(s_{t+1} | s_t, a_t) P_{t+1}^Z(z_{t+1} | s_{t+1}) \widehat{V}_{t+1,rect}^{\pi, RUDB}(b_{t+1}) + \\
&\quad \underbrace{\sum_{\tau_{t:t+1} \notin \widehat{\mathcal{T}}_{t:t+1}} b_t(s_t) P_t^T(s_{t+1} | s_t, a_t) P_{t+1}^Z(z_{t+1} | s_{t+1}) \widehat{V}_{t+1,rect}^{\pi, RUDB}(b_{t+1})}_{\text{zero according to (91)}} \\
&\quad \equiv 0
\end{aligned} \tag{93}$$

on the sampled support the models P_t^T, P_{t+1}^Z equal their projected counterparts $\widehat{P}_t^T, \widehat{P}_{t+1}^Z$:

$$= \sum_{\tau_{t:t+1} \in \widehat{\mathcal{T}}_{t:t+1}} b(s_t) \widehat{P}_t^T(s_{t+1} | s_t, a_t) \widehat{P}_{t+1}^Z(z_{t+1} | s_{t+1}) \widehat{V}_{t+1,rect}^{\pi, RUDB}(b_{t+1}) \tag{94}$$

$$= \sum_{z_{t+1} \in \widehat{\mathcal{Z}}_{t+1}} \widehat{\text{Pr}}(z_{t+1} | b_t, a_t) \widehat{V}_{t+1,rect}^{\pi, RUDB}(b_{t+1}) \tag{95}$$

$$= \mathbb{E}_{\widehat{P}_t^T, \widehat{P}_{t+1}^Z}^{\pi, RUDB} \left[\widehat{V}_{t+1,rect}^{\pi, RUDB}(b_{t+1}) \right] \tag{96}$$

Moving on to the **right** expectation in (89) we now show that: $\mathbb{E} \left[\epsilon_{t+1,rect}^{\pi, RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) \right] \leq \epsilon_{t,rect}^{\pi, RUDB}(b_t, a_t)$. The intuition is that the parent error $\epsilon_{t,rect}^{\pi}$ is greater than the expectation over the child error, since parent is worst-case over **current**+future models choices, while the child expectation is over a **fixed current** models:

$$\begin{aligned}
\mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} \left[\epsilon_{t+1,rect}^{\pi, RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) \right] &= \sum_{z_{t+1}} \text{Pr}(z_{t+1} | b_t, a_t) \epsilon_{t+1,rect}^{\pi, RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) \\
&= \sum_{z_{t+1} \in \widehat{\mathcal{Z}}_{t+1}} \widehat{\text{Pr}}(z_{t+1} | b_t, a_t) \epsilon_{t+1,rect}^{\pi, RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})).
\end{aligned} \tag{97}$$

To justify the last transition, we restate here the action-value error term defined in (75):

$$\epsilon_{t+1,rect}^{\pi, RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) = R_{max} \left(1 - \underbrace{\sum_{z_{t+2} \in \widehat{\mathcal{Z}}_{t+2}} \widehat{\text{Pr}}^*(z_{t+2} | b_{t+1}, \pi_{t+1}(b_{t+1}))}_{(i)} \right) + \underbrace{\mathbb{E}_{z_{t+2}|b_{t+1}, \pi_{t+1}}^{\widehat{P}_{t+1}^{*T}, \widehat{P}_{t+2}^{*Z}} \left[\epsilon_{t+2,rect}^{\pi, RUDB}(b_{t+2}) \right]}_{(ii)}, \tag{98}$$

(ii) Is independent of z_{t+1} , thus in (97) $\sum_{z_{t+1}} \text{Pr}(z_{t+1} | b_t, a_t) \cdot \mathbb{E}[\cdot] = \mathbb{E}[\cdot]$.

(i) Does depend on z_{t+1} , and is nonzero only on the sampled trajectory $\widehat{\mathcal{T}}_{t,j}$. As discussed above in (91), which justifies the last transition.

By Adding the nonnegative term $R_{max} \left(1 - \sum_{z_{t+1}} \widehat{\text{Pr}}(z_{t+1}) \right)$ to (97), we get the following inequality:

$$R_{max} \left(1 - \sum_{z_{t+1}} \widehat{\text{Pr}}(z_{t+1} | b_t, a_t) \right) + \sum_{z_{t+1} \in \widehat{\mathcal{Z}}_{t+1}} \widehat{\text{Pr}}(z_{t+1} | b_t, a_t) \epsilon_{t+1,rect}^{\pi, RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) \tag{99}$$

$$\leq \epsilon_{t,rect}^{\pi, \widehat{P}_t^T, \widehat{P}_{t+1}^Z}(b_t, a_t), \tag{100}$$

$\epsilon_{t,rect}^{\pi, \widehat{P}_t^T, \widehat{P}_{t+1}^Z}(b_t, a_t)$ denotes the child error for fixed current models $\widehat{P}_t^T, \widehat{P}_{t+1}^Z$. Referring back to (89), we have shown that:

$$= r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \left\{ \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} \left[\widehat{V}_{t+1,rect}^{\pi, RUDB}(b_{t+1}) \right] + \mathbb{E}_{z_{t+1}|b_t, a_t}^{P_t^T, P_{t+1}^Z} \left[\epsilon_{t+1,rect}^{\pi, RUDB}(b_{t+1}, \pi_{t+1}(b_{t+1})) \right] \right\} \tag{101}$$

$$\leq r(b_t, a_t) + \inf_{\substack{P_t^T \in \mathcal{P}_t^T \\ P_{t+1}^Z \in \mathcal{P}_{t+1}^Z}} \left\{ \mathbb{E}_{z_{t+1}|b_t, a_t}^{\widehat{P}_t^T, \widehat{P}_{t+1}^Z} \left[\widehat{V}_{t+1,rect}^{\pi, RUDB}(b_{t+1}) \right] + \epsilon_{t,rect}^{\pi, \widehat{P}_t^T, \widehat{P}_{t+1}^Z}(b_t, a_t) \right\}. \tag{102}$$

We use Lemma 2, which shows that the last infimum can be over the projected uncertainty sets, namely:

$$\inf_{P \in \mathcal{P}} \widehat{f}(\mathcal{R}(P)) = \inf_{\widehat{P} \in \widehat{\mathcal{P}}} \widehat{f}(\widehat{P}), \quad (103)$$

Therefore

$$= r(b_t, a_t) + \inf_{\substack{\widehat{P}_t^T \in \widehat{\mathcal{P}}_t^T \\ \widehat{P}_{t+1}^Z \in \widehat{\mathcal{P}}_{t+1}^Z}} \left\{ \widehat{\mathbb{E}}_{z_{t+1}|b_t, a_t}^{\widehat{P}_t^T, \widehat{P}_{t+1}^Z} \left[\widehat{V}_{t+1, rect}^{\pi, RUDB}(b_{t+1}) \right] + \epsilon_{t, rect}^{\pi, \widehat{P}_t^T, \widehat{P}_{t+1}^Z}(b_t, a_t) \right\} \quad (104)$$

$$\triangleq RUDB^\pi(b_t, a_t), \quad (105)$$

which concludes the proof. $\square$